

Open Climate Data on a Globe

The background knowledge behind the *earth* project: GIBS, tiles, colormaps, product levels, climatologies and the datasets in the app

This primer collects, in one place, the concepts needed to understand how open climate data travels from a satellite or a rain gauge onto an interactive 3-D globe — and why the app is built the way it is. It is written for a technically-minded reader; no remote-sensing background is assumed.

Companion documents in the repository: **docs/CATALOG.md** (241+ dataset catalog), **docs/COMBINING_DATASETS.md** (which datasets measure the same quantity and how they may be combined), **docs/SPECIES_AND_CLIMATE.md**.

1 · What is GIBS?

GIBS — the **Global Imagery Browse Services** — is NASA's public tile server for satellite imagery. It serves over 1,000 visualised data products (true-colour mosaics, sea-surface temperature, precipitation, snow cover, aerosols, night lights, ...) as small pre-rendered map tiles that any web map client can request over plain HTTPS, with no API key or account. It is the engine behind NASA Worldview, and it is where most of the raster layers on the earth globe come from.

The crucial idea is that GIBS serves **pictures, not numbers**. A GIBS tile is an ordinary PNG or JPEG in which each data value has been mapped through a published colour table. That makes display trivially fast — the browser just draws images — but it means that recovering the physical value at a pixel requires inverting the colormap (section 4), and that any computation (differences, averages) must happen on the client after that inversion. The alternative — downloading the underlying NetCDF/HDF granules and rendering from raw numbers — is more exact but orders of magnitude heavier; GIBS is the pragmatic middle ground that makes a globe of daily global imagery feasible in a browser.

A GIBS tile URL is fully deterministic:

```
https://gibs.earthdata.nasa.gov/wmts/epsg4326/best/{Layer}/default/{Time}/{TileMatrixSet}/{Zoom}/{Row}/{Col}.png
```

Layer is an identifier such as `GHRSSST_L4_MUR_Sea_Surface_Temperature`. **Time** is an ISO date — GIBS layers are time-dimensioned, so the same URL pattern reaches two decades of daily history. **best** means "the best available version of each date" (near-real-time at first, replaced by the final science product when it arrives, typically days later). The **GetCapabilities** XML document lists every layer with its valid time ranges, tile matrix set, format, and colormap — the app reads it to know what exists.

2 · Tiles, WMTS and the map pyramid

WMTS (Web Map Tile Service) is the OGC standard GIBS implements. The world is cut into a pyramid of square tiles: at each zoom level the map is divided into a grid twice as fine as the level above, and a client fetches only the handful of tiles its viewport touches. This is the same mechanism behind every slippery web map since Google Maps.

GIBS's geographic (EPSG:4326) pyramid has a quirk worth knowing: it does *not* start from one world-spanning tile. Level 0 is already **2×1** tiles, level 1 is **3×2**, and the resolution per level is $0.5625/2^L$ degrees per pixel with 512-pixel tiles. Because 3 is not a power of two, the right-most and bottom-most tiles at some levels extend past the edge of the map — they are *partial* tiles. A subtle but critical rule: a tiling scheme must declare the **full nominal extent** of such tiles, not the clamped visible portion, or the renderer maps pixels to the wrong longitudes. (Getting this wrong blanked the Pacific in an early version of the app — hence the custom `GIBSGeographicTilingScheme` and a regression test pinned to the published matrix definitions.)

Projections. EPSG:4326 ("geographic") maps longitude/latitude directly to x/y — natural for draping onto a 3-D globe. Most 2-D web maps instead use EPSG:3857 (Web Mercator), which GIBS also offers. National products often use a local projection — MeteoSwiss grids are delivered in the Swiss LV95 metre grid, with per-cell longitude/latitude arrays included so consumers can resample without a projection library.

3 · How a measurement becomes a map: product levels

Satellite data is published in **processing levels**, and knowing them explains most of what you see:

Level	What it is	Example in the app
L0–L1	Raw instrument counts; calibrated, geolocated radiances.	—
L2	A geophysical quantity along the satellite's swath. Gaps between orbits, stripes, cloud masks.	MODIS chlorophyll (visible swath seams)
L3	Swaths binned onto a regular global grid for a day / 8 days / a month.	SMAP salinity monthly, MODIS L3 SST
L4	A gap-free <i>analysis</i> : observations from one or more sensors blended with interpolation or a model.	MUR SST 1 km (the default layer)

Two practical consequences. First, **latency**: daily products appear hours to days after observation (the app defaults to "two days ago" for safety), and **monthly composites only exist for completed months** — requesting the current month returns nothing, which is why the salinity layer snaps back to the last complete month. Second, **gaps are real**: instruments fail and missions age. SMAP salinity has a 2024 data gap; AMSR2 sea ice lags its mission's availability windows. A blank layer is more often a data-availability fact than a bug.

4 · Colormaps, legends, and reading values back

Every colormapped GIBS layer publishes an XML colour table mapping RGB triples to physical value ranges (e.g. `rgb="255,0,0" value="[30,31]"` Kelvin). The app uses these tables three ways: to draw **interactive legends** (hover a colour, see its value); to run the **value probe** (click the globe → sample the rendered pixel → invert RGB back to a value); and to power **computed differences** — invert both dates to numbers per pixel, subtract, and re-colour on a diverging blue↔red scale.

Inversion has limits worth understanding. It recovers the *bin centre*, not the exact value — a colormap with 250 bins over 40 °C quantises to ≈ 0.16 °C. It only works where the mapping is one-to-one and the field is **continuous** (SST, sea-ice fraction, land temperature). It fails conceptually for **instantaneous, sparse fields**: precipitation rate is log-scaled, transparent below 0.1 mm/hr, and mostly "no rain right now" — differencing two snapshots measures whether it happened to be raining at overpass time, not a climate signal. That is why the app offers differences for continuous rasters only, and answers "how has rainfall changed?" with climatologies instead (next section).

5 · Instantaneous fields, anomalies and climatologies

Climate datasets answer three different questions, and mixing them up is the most common source of confusion:

Kind	Question it answers	Examples in the app
Instantaneous / daily state	What is happening <i>now</i> / <i>on date X</i> ?	IMERG rain rate, MUR SST, sea ice, chlorophyll, salinity
Anomaly	How does date X differ from the long-term average for that place and season?	SST anomalies (MUR25); GISTEMP series (vs 1951–1980)
Climatology / normal	What is <i>typical</i> here, averaged over decades?	GPCP precip (global), E-OBS (Europe), MeteoSwiss normal 1991–2020 (Swiss), OISST 1991–2020 SST mean

An anomaly is always *relative to a stated baseline* — GISTEMP's +1.3 °C is "versus the 1951–1980 mean", the WMO's current normal period is 1991–2020, and comparing numbers across different baselines silently shifts them. A climatology, conversely, deliberately hides weather: GPCP's Amazon cell says $\sim 2,300$ mm/yr falls there *on average*, not whether it rained yesterday. The app pairs each live field

with its baseline where possible: MUR (today's SST) sits alongside OISST (the 30-year mean it is judged against); IMERG (rain now) alongside GPCP/E-OBS/MeteoSwiss (rain normally).

6 · Beyond tiles: how the non-GIBS data gets in

Not everything is a picture on a tile server. The rest of the app's data arrives through a handful of standard access patterns worth recognising across the field:

Pattern	What it is	Used for
NetCDF / HDF	Self-describing binary arrays with named dimensions and units; the lingua franca of climate science.	GPCP, E-OBS, OISST, MeteoSwiss grids; RAPID transport series
OPeNDAP / THREDDS	Protocols for subsetting remote NetCDF without downloading whole files.	OC-CCI ocean colour (catalogued)
ERDDAP	NOAA's data server; slices many datasets into CSV/JSON via URL queries.	Argo float positions
STAC	SpatioTemporal Asset Catalog — JSON catalogs of assets with bounding boxes and times.	MeteoSwiss open data (data.geo.admin.ch)
REST APIs	Ordinary JSON web services.	Climate TRACE emitters, GBIF occurrences
Bulk archives	Tarballs / parquet on plain HTTP.	RGI v7 glacier outlines, Hugonnet 2021 elevation-change rates

The app converts each of these once, offline, into small static JSON snapshots (`scripts/refresh_data.py`), so the browser never depends on a third-party API being up. Grids with no tile service are painted client-side by a small canvas renderer (`GridProvider`) from those snapshots.

7 · The app's layers at a glance

Layer	Source & type	Cadence / period	Resolution
True colour	VIIRS mosaic (GIBS tiles)	daily, ~3 h latency	250 m
Sea surface temperature	MUR L4 analysis (GIBS)	daily, 2002–	1 km
SST anomalies	MUR25 L4 (GIBS)	daily, 2002–	25 km
SST climatology	NOAA OISST v2.1 (grid)	1991–2020 mean	0.25°→1°
Precipitation rate	GPM IMERG V07 (GIBS)	daily + 30-min, 2000–	~10 km
Precip climatology (global)	GPCP v2.3 (grid)	mean annual, 1979–	2.5°
Precip climatology (Europe)	E-OBS v31 (grid)	mean annual, 1950–	0.25°
Precip normal (Switzerland)	MeteoSwiss OGD (grid)	1991–2020 normal	~2 km
Sea ice concentration	AMSR2 (GIBS)	daily, 2012–	12 km
Snow cover	MODIS NDSI (GIBS)	daily, 2000–	500 m
Land surface temperature	MODIS Terra (GIBS)	daily	1 km
Aerosol optical depth	MODIS combined (GIBS)	daily, 2017–	10 km
Chlorophyll-a	PACE/OCI, NASA Ocean Color (GIBS)	daily, 2024–	1 km
Sea surface salinity	SMAP L3 (GIBS)	monthly, 2015– (2024 gap)	~60 km
Night lights	VIIRS Black Marble (GIBS)	annual composite	500 m
Glaciers	RGI v7 + Hugonnet 2021 (points)	~2000 outlines; 2000–2020 dh/dt	274k glaciers
Emitters / floats / species	Climate TRACE · Argo · GBIF (points)	snapshots	—

8 · Small glossary

Term	Meaning
WMTS / WMS	OGC standards for serving maps as pre-cut tiles (WMTS) or on-demand rendered images (WMS).
EPSG code	Numeric identifier of a coordinate reference system (4326 = lon/lat; 3857 = Web Mercator; 2056 = Swiss LV95).
Granule	One file of satellite data — typically one orbit segment or one day-tile of a product.
Analysis (L4)	A gap-free gridded field produced by blending observations, e.g. with optimal interpolation (the OI in OISST).
Climatology / normal	A multi-decade average; WMO normals span 30 years, currently 1991–2020.
Anomaly	Departure of an observation from a stated climatological baseline.
Composite	A product built by aggregating a period (8-day, monthly); exists only once the period is complete.
NRT vs science quality	Near-real-time products arrive in hours but are later replaced by re-processed, better-calibrated versions — GIBS's 'best' endpoint always shows the best one available.
Ensemble	Several independent estimates of the same field; their spread is a measure of structural uncertainty (the app's SST ensemble layer).
Geodetic mass balance	Glacier change measured by differencing elevation models over time (Hugonnet 2021's dh/dt, m/yr).